Voltmeter



Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Voltmeter

What it is used for

A voltmeter measures the electrical potential difference between two points in a circuit. Correct meter settings and parallel connection help obtain reliable readings while reducing the risk of electric shock and instrument damage.

Main parts

- Display or scale shows the measured voltage.
- Selector switch chooses voltage type and measurement range.
- Positive input terminal accepts the red test lead.
- COM terminal accepts the black common test lead.
- Test leads and probes connect the meter to the circuit.
- Internal protection helps protect the meter within its rated limits.
- Battery supplies power to digital models.

Before you start

1. Read the meter instructions and confirm its voltage rating suits the circuit.
2. Inspect the meter, probes, insulation, leads, and terminals for damage.
3. Select AC or DC voltage according to the circuit being tested.
4. Start with a range above the expected voltage when using a manual-range meter.
5. Insert the black lead into COM and the red lead into the voltage terminal.
6. Keep hands behind the probe finger guards and wear appropriate PPE for the task.

How to use

1. Switch off circuit power before connecting probes whenever the task permits.
2. Set the selector to the correct voltage type and suitable range.
3. Connect the probes across the two points being measured, in parallel with the circuit component.
4. For DC measurements, observe polarity and connect red to the more positive point when known.
5. Restore power only when the connections are secure and the procedure permits live measurement.
6. Read the display or scale without touching exposed conductive parts.
7. If the reading exceeds the selected range, disconnect safely and select a higher range.
8. Record the reading with the unit and whether it is AC or DC.
9. Switch off the circuit before removing probes when possible.
10. Remove the leads, switch off the meter, and store it safely.

⚠ Safety

- Never connect a voltmeter in series as a current-measuring device.
- Do not exceed the meter's voltage or measurement-category rating.
- Never use damaged probes, exposed wires, or cracked meter insulation.
- Avoid touching metal probe tips or live conductive parts during measurement.
- Use appropriate isolation and authorised supervision when testing mains or high-voltage circuits.
- Never change exposed wiring or terminals while the circuit is energised.

Care & storage

- Disconnect the leads and switch off the meter before cleaning it.
- Wipe the case with a dry or manufacturer-approved cloth.
- Store probes without sharp bends and protect their insulation from damage.
- Check the battery, leads, and calibration status according to the manual.

Common mistakes to avoid

- Selecting AC when measuring DC, or selecting DC when measuring AC.
- Connecting the meter in series instead of parallel.
- Using a range below the expected voltage on a manual-range meter.
- Touching exposed probe tips or using damaged test leads.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.